

Algebra: Chapter 0, Aluffi

Part III: Rings and Modules

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Notation:

- Given a ring R and element $a \in R$, I denote by $\langle a \rangle$ the principal *ideal* generated by a in the ring R and $\langle a \rangle^+$ the *subgroup* generated by a in the underlying additive group R^+ .

§3 Ideals and Quotient Rings

4.

Let $\varphi : \mathbb{Z} \rightarrow R$ be the unique ring homomorphism from $\mathbb{Z}$ to R . By definition of a ring homomorphism, $\varphi(1_{\mathbb{Z}}) = 1_R$. By assumption on R , the subgroup $\langle 1_R \rangle^+$ of the additive group R^+ is equal to the ideal $\langle 1_R \rangle$ of R . $\langle 1_R \rangle = R$, of course. Also, notice $\text{im } \varphi = \langle 1_R \rangle^+$. Hence $\text{im } \varphi = R$.

By definition, $n := \text{char } R$ is the kernel of $\varphi : \mathbb{Z} \rightarrow R$, hence the first isomorphism theorem yields

$$\mathbb{Z}/n\mathbb{Z} \cong R.$$

5.

Let $e_{ij} \in \mathcal{M}_n(R)$ denote the matrix with a 1 on the (i, j) th entry and 0's everywhere else. Suppose $A \in J$. Since J is a two-sided ideal,

$$e_{rs}Ae_{kl} \in J, \quad (\forall r, s, k, l \in \{1, \dots, n\})$$

What is this the matrix $e_{rs}Ae_{kl}$? Let $A = [a_{ij}]$. Writing $[a'_{ij}] := Ae_{kl}$,

$$\begin{aligned} a'_{ij} &= \sum_{u=1}^n a_{iu}(e_{kl})_{uj} \\ &= \begin{cases} 0, & \text{if } j \neq l \\ a_{ik}, & \text{if } j = l. \end{cases} \end{aligned}$$

Next, writing $[a''_{ij}] = e_{rs}[a'_{ij}]$,

$$\begin{aligned} a''_{ij} &= \sum_{u=1}^n (e_{rs})_{iu} a'_{uj} \\ &= \begin{cases} 0, & \text{if } j \neq l \text{ or } i \neq r \\ a'_{sj} = a_{sk}, & \text{if } j = l \text{ and } i = r \end{cases} \end{aligned}$$

This shows that $e_{rs}Ae_{kl}$ is the matrix whose (r, l) th entry is the (s, k) th entry of A and has 0's everywhere else. This proves the forward direction.

Next, suppose every entry of a of A , there is a matrix $ae_{ij} \in J$ for all $i, j \in \{1, \dots, n\}$. Then clearly $A \in J$: writing $A = [a_{ij}]$,

$$A = \sum_{1 \leq i, j \leq n} a_{ij} e_{ij}.$$

and because each $a_{ij}e_{ij} \in J$, so is their sum, i.e., $A \in J$.

12.

The nilradical of a commutative ring is an ideal because the binomial theorem holds in commutative rings.

$\mathcal{M}_n(\mathbb{R})$ for $n > 1$ is a noncommutative ring whose set of nilpotent elements is not an ideal. Recall that lower/upper triangular matrices with 0's on the diagonal are nilpotent. However, the sums of such matrices may be an invertible matrix; the powers of an invertible matrix are of course cannot be the zero matrix. For example, in $\mathbb{R}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ are nilpotent. However, their sum is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, which is an invertible matrix. This demonstrates that the set of nilpotent elements of $\mathcal{M}_2(\mathbb{R})$ does not form an ideal.

14.

Suppose that $\text{char} R = nm$ for integers $n, m > 1$. In R , let $n := \underbrace{1 + \dots + 1}_{n \text{ times}}$ and $m := \underbrace{1 + \dots + 1}_{m \text{ times}}$. Then by the assumption on the character of R ,

$$n, m \neq 0, \text{ but } nm = 0,$$

which violates the fact that R is an integral domain. Hence the character of an integral domain is either 0 or a prime integer.

Clearly, the zero ring is the only ring with character 1.

15.

$1 + 1 = (1 + 1)(1 + 1) = 1 + 1 + 1 + 1$ shows that $1 + 1 = 0$ since addition can be cancelled in any ring; the character of a boolean ring is 2. Next, for all $a, b \in R$,

$$\begin{aligned} a + b &= (a + b)^2 \\ &= a^2 + ab + ba + b^2 \\ &= a + b + ab + ba, \end{aligned}$$

so $ab + ba = 0$, i.e., $ab = -ba$. Because R has character 2, $ba = -ba$, from which follows $ab = ba$; R is commutative.

Suppose any boolean ring is an integral domain. Then for any $a \neq 0$, the cancellation law applies to the identity $a^2 = a$ to show that $a = 1$. Hence any boolean ring that is an integral domain is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

§4 Ideals and Quotients: Remarks and Examples

9.

Since $f(x)$ is a zerodivisor, choose $g(x) \in R[x]$ of minimal degree such that $fg = 0$. Write $f(x) = a_n x^n + \dots + a_0$ and $g(x) = b_m x^m + \dots + b_0$ with coefficients in R . Also, let $f(x)g(x) = c_{n+m} x^{n+m} + \dots + 0$, where each $c_t = 0$ by assumption. I claim that each a_i for $i \in \{0, \dots, n\}$ annihilates g , that is,

$$a_i g(x) = 0 \quad (i \in \{0, \dots, n\})$$

Induct on the i , starting from $i = n$. Using the regular rules for polynomial multiplication,

$$c_{n+m} = a_n b_m,$$

and $c_{n+m} = 0$ by assumption. Hence $a_n b_m = 0$. Because R is commutative, $f(a_n g(x)) = a_n f(x)g(x) = 0$, and $a_n b_m = 0$ means $a_n g(x)$ is a polynomial of lower degree than g . The minimality of g thus forces

$$a_n g(x) = 0;$$

a_n annihilates g .

Next, assume $a_n, \dots, a_{n-k+1}$ annihilate g . Once again, starting from the assumption that $c_{n+m-k} = 0$ and using the regular rules for polynomial multiplication,

$$\begin{aligned} 0 = c_{n+m-k} &= \sum_{i=n-k}^n a_i b_{n+m-k-i} \\ &= a_{n-k} b_m, \end{aligned}$$

where the last inequality follows from the inductive hypothesis. As in the base case, $a_{n-k}g(x)$ is a polynomial of lower degree than $g(x)$ such that $f(x)(a_{n-k}g(x)) = 0$, so one concludes $a_{n-k}g(x) = 0$.

Having shown that each coefficient a_i annihilates g , it is clear that $b_j f(x) = 0$ for each $j \in \{0, \dots, m\}$. Namely, $b_0 f(x) = 0$, as was to be shown (this also shows that $g(x) = b_0$, i.e., a constant polynomial).

11.

(a)

Define the substitution map

$$\varphi : R[x] \rightarrow R; \quad \varphi_R = \text{id}, \quad x \mapsto a.$$

By Prop 4.6 in the book, $\ker \varphi = \langle x - a \rangle$ and φ is surjective. I now freely use a fact that technically isn't covered yet in the book: since $x - a$ is an irreducible polynomial in $R[x]$,

$$R[x]/\langle x - a \rangle \cong R$$

as rings, not just as abelian groups. $I := \langle f_1(x), \dots, f_r(x), x - a \rangle$ contains $\ker \varphi$, so $J := \varphi(I)$ is an ideal in $R/\langle x - a \rangle$. Namely, J is generated by the images of each of the generators of I under φ ;

$$J = \langle \overline{f_1(a)}, \dots, \overline{f_r(a)} \rangle.$$

By the correspondence theorem, $\varphi^{-1}(J) = I$, but one also sees $\varphi^{-1}(J) = \langle f_1(a), \dots, f_r(a), x - a \rangle$ just by taking preimage of the generators of J under φ . It follows that $\langle f_1(x), \dots, f_r(x), x - a \rangle = \langle f_1(a), \dots, f_r(a), x - a \rangle$.

(b)

Letting bars denote images under the substitution homomorphism φ as defined above,

$$\begin{aligned} \frac{R[x]}{\langle f_1(x), \dots, f_r(x), x-a \rangle} &\cong \frac{R[x]/\langle x-a \rangle}{\langle \overline{f_1(x)}, \dots, \overline{f_r(x)}, \overline{x-a} \rangle} \\ &\cong \frac{R}{\langle \overline{f_1(a)}, \dots, \overline{f_r(b)} \rangle} \end{aligned}$$

where the last equivalence, particularly in the denominator, follows because the image of $f_i(x)$ under φ is just substitution by $x = a$.

12.

Induct on n . $n = 1$ has already been proven; $R[x]/\langle x-a \rangle \cong R$ as rings. Suppose the proposition is true for $n-1$. Consider the ring homomorphism defined by

$$\varphi : R[x_1, \dots, x_n] \rightarrow R[x_1, \dots, x_{n-1}]; \quad \varphi_{R[x_1, \dots, x_{n-1}]} = \text{id}, \quad x_n \mapsto a_n.$$

Viewing $R[x_1, \dots, x_n]$ as $R[x_1, \dots, x_{n-1}][x_n]$, i.e., polynomials with coefficients in $R[x_1, \dots, x_{n-1}]$ with indeterminate x_n , example 4.7 gives $\ker \varphi = \langle x_n - a_n \rangle$. By the first isomorphism theorem and prop 3.11 in the book,

$$\begin{aligned} \frac{R[x_1, \dots, x_n]}{\langle x_1 - a_1, \dots, x_n - a_n \rangle} &\cong \frac{R[x_1, \dots, x_n]/\langle x_n - a_n \rangle}{\langle \overline{x_1 - a_1}, \dots, \overline{x_{n-1} - a_{n-1}}, \overline{x_n - a_n} \rangle} \\ &\cong \frac{R[x_1, \dots, x_{n-1}]}{\langle \overline{x_1 - a_1}, \dots, \overline{x_{n-1} - a_{n-1}} \rangle} \end{aligned}$$

By the inductive hypothesis, this last quotient ring is isomorphic to R .

13.

Take any $k \in \{1, \dots, n\}$. By Question 12,

$$\frac{R[x_1, \dots, x_n]}{\langle x_1, \dots, x_k \rangle} \cong R[x_{k+1}, \dots, x_n]$$

where the latter is an integral domain because R is an integral domain. Hence $\langle x_1, \dots, x_k \rangle$ is a prime ideal.

14.

Take maximal ideal $M \triangleleft R$ and $ab \in M$. Suppose $a \notin M$. Since M is maximal, $M + \langle a \rangle = R$, hence $1 = m + ad$ for some $m \in M$ and $d \in R$. Then

$$b = b(m + ad) = mb + abd \in M,$$

as was to be shown.

§5 Modules Over a Ring

9.

Define

$$\begin{aligned}\Phi : R &\rightarrow \text{End}_{R\text{-Mod}}(M); \quad r \mapsto \sigma_r \\ \sigma_r : M &\rightarrow M; \quad m \mapsto rm\end{aligned}$$

σ_r is clearly an R -module homomorphism. Further, for all $r, s, a \in R$ and all $m \in M$,

$$\begin{aligned}\Phi(r + as)(m) &= \sigma_{r+as}(m) \\ &= (r + as)(m) = rm + asm \\ &= \sigma_r(m) + \sigma_a\sigma_s(m) = (\Phi(r) + \Phi(a)\Phi(s))(m),\end{aligned}$$

hence Φ is a ring homomorphism. It is clear that $\text{im } \Phi \subset Z(\text{End}_{R\text{-Mod}}(M))$. This shows that $\text{End}_{R\text{-Mod}}(M)$ is an R -algebra. $(\text{End}_{R\text{-Mod}}(M), \Phi)$ is a natural R -algebra in the sense that Φ induces the following action of R on $\text{End}_{R\text{-Mod}}(M)$:

$$\rho : R \times \text{End}_{R\text{-Mod}}(M) \rightarrow \text{End}_{R\text{-Mod}}(M); \quad (r, \varphi) \mapsto r\varphi,$$

where $r\varphi : M \rightarrow M$ is the R -module homomorphism defined with respect to r and φ as usual.

$\mathcal{M}_n(R)$ is also an R -algebra, given by the homomorphism

$$\Phi : R \rightarrow \mathcal{M}_n(R); \quad r \mapsto \begin{bmatrix} r & & & \\ & r & & \\ & & \ddots & \\ & & & r \end{bmatrix},$$

i.e., realize $r \in R$ as the $n \times n$ scalar matrix. Scalar matrices clearly commute with all other matrices in $\mathcal{M}_n(R)$.

11.

First, take a R -module homomorphism $\varphi : M \rightarrow M$. Then define an $R[x]$ -module structure on M as follows: given $f(x) = a_n x^n + \dots + a_0 \in R[x]$,

$$\rho : R[x] \times M \rightarrow M; \quad (f(x), m) \mapsto a_n \varphi^n(m) + a_{n-1} \varphi^{n-1}(m) + \dots + a_0 m,$$

i.e., the $R[x]$ -module structure of M is fully determined by the underlying R -module structure of M and the mapping $\rho(x, m) = \varphi(m)$ since, by definition, $R[x]$ is generated by R and x . Instead of expliciting writing ρ , I just use multiplication notation for the following verification that this does indeed define an $R[x]$ -module structure on M . It is clear that for any $f(x), g(x) \in R[x]$ and any $m_1, m_2 \in M$, $(f(x) + g(x))(m_1) = f(x)m_1 + g(x)m_1$ and $f(x)(m_1 + m_2) = f(x)m_1 + f(x)m_2$. Write $f(x) = a_n x^n + \dots + a_0$ and $g(x) = b_m x^m + \dots + b_0$.

Without loss of generality, assume $n = m$.

$$\begin{aligned}
 (f(x)g(x))m &= \left(\sum_{1 \leq i, j \leq n} a_i b_j x^{i+j} \right) m \\
 &= \sum_{1 \leq i, j \leq n} a_i b_j \varphi^{i+j}(m) \\
 &= \sum_{i=1}^n a_i \varphi^i \left(\sum_{j=1}^n b_j \varphi^j(m) \right) \\
 &= f(x)(g(x)m)
 \end{aligned}$$

where the third equality follows from the fact that φ is a R -module homomorphism. This shows that ρ is defines a $R[x]$ -module structure on M .

Vice versa, take any action $\rho : R[x] \times M \rightarrow M$ that defines an $R[x]$ -module structure on M . Then one can easily check that the map

$$\varphi : M \rightarrow M; \quad m \mapsto \rho(x, m)$$

is an R -module homomorphism. This establishes the desired bijection.

13.

Let $I = \langle a \rangle$ as an ideal in R for some $a \in R \setminus 0$.

$$\varphi : R \rightarrow I; \quad r \mapsto ra$$

is an R -module homomorphism: for all $r_1, r_2, c \in R$, $\varphi(r_1 + cr_2) = (r_1 + cr_2)a = r_1a + cr_2a = \varphi(r_1) + c\varphi(r_2)$. By definition, φ is surjective. It is also injective because R is an integral domain: since $a \neq 0$, $ra = 0 \iff r = 0$. This shows that $I \cong R$ as R -modules.

15.

This is a confusing problem, even though, in the end, there's not much to it. Consider

$$\varphi : I \rightarrow I \cdot (R/J); \quad a \mapsto a + J.$$

First, notice that this definition does indeed make sense because $I \cdot (R/J) \triangleleft R/J$ as a submodule. For any $a, b \in I$ and any $r \in R$, the computation $\varphi(a + rb) = (a + rb) + J = (a + J) + r(b + J) = \varphi(a) + r\varphi(b)$ shows that φ is an R -module homomorphism. φ is surjective because elements of $I \cdot (R/J)$ are of the form

$$\sum a_i(r_i + J) = \sum (a'_i + J) \quad (a_i, a'_i \in I, r_i \in R)$$

where $a'_i = a_i r_i \in I$ since I is an ideal in R . It is clear that $\ker \varphi = I \cap J$, hence

$$I/(I \cap J) \cong I \cdot (R/J)$$

as R -modules. By the second isomorphism theorem for modules, $(I+J)/J \cong I/(I \cap J)$, so $(I+J)/J \cong I \cdot (R/J)$.

16.

Clearly, if $M = 0$, then $aM = 0$ as well, hence $aM = M$. Conversely, suppose $aM = M$. Let $a^n = 0$. For any $m \in M$, $\exists m_1 \in M$ such that $m = am_1$. But for this m_1 , $\exists m_2 \in M$ such that $m_1 = am_2$, i.e., $m = a^2m_2$. Repeating this argument, there is an $m_n \in M$ such that $m = a^n m_n$. But $a^n = 0$, hence $m = 0$. This shows that $M = 0$.

§6 Products, Coproducts, etc. in $R\text{-Mod}$

3.

The difficulty with this question is knowing how to map $M \rightarrow \ker p$. Thankfully, one has inspirations from projections in linear algebra that maps a vector to its “projection” into a fixed subspace. Given any vector v , one can use this projection map, call it p , to determine the component of v that is orthogonal to the subspace: it is given by $v - p(v)$. Of course, projection of vectors orthogonal to the subspace is just 0, i.e., the kernel of the projection map is precisely vectors orthogonal to the subspace.

Something similar can be done here:

$$o : M \rightarrow \ker p; \quad m \mapsto m - p(m) \quad (1)$$

gives a well-defined mapping, and in fact it is an R -module homomorphism because p is so. (1) and

$$p' : M \mapsto \operatorname{im} p; \quad m \mapsto p(m) \quad (2)$$

can be used to demonstrate the desired R -module isomorphism $M \cong \ker p \oplus \operatorname{im} p$. The direct way to show this is checking

$$\varphi : M \mapsto \ker p \oplus \operatorname{im} p; \quad m \mapsto (o(m), p(m))$$

is an R -module isomorphism; the only notable point in this procedure is checking the surjectivity of φ : for any $(n, p(m)) \in \ker p \oplus \operatorname{im} p$, $\varphi(p(m) + n) = (n, p(m))$.

Alternatively, verify that M with the mappings (1) and (2) satisfies the universal property for the product of $\operatorname{im} p$ and $\ker p$: indeed, in the category of R -modules, suppose the diagram below commutes for some $\sigma : K \rightarrow M$:

$$\begin{array}{ccc}
 & & \operatorname{im} p \\
 & \nearrow f & \\
 K & \xrightarrow{\sigma} & M \\
 & \searrow o & \\
 & & \ker p
 \end{array}$$

(Note: In the original image, there are also curved arrows from K to $\operatorname{im} p$ labeled f and from K to $\ker p$ labeled g . The diagram above represents the commutative triangle with straight arrows.)

The commutativity of the diagram forces $f(k) = p'(\sigma(k))$ and $g(k) = \sigma(k) - p(\sigma(k))$ for all $k \in K$. Combining these two equations yields $\sigma(k) = f(k) + g(k)$ for all $k \in K$, thus giving a unique definition for σ . And indeed, with this definition, $\sigma : K \rightarrow M$ is an R -module homomorphism because f and g are so. This verifies $M \cong \ker p \oplus \operatorname{im} p$.

6.

This question has taught me to stop relying on category theory for intuition. The key point in this question is finding n R -mod homomorphisms $R^{\oplus n} \rightarrow R$ that span $\text{Hom}_R(R^{\oplus n}, R)$. We know exactly the set of n “natural” morphisms $R^{\oplus n} \rightarrow R$: the coordinate projections

$$\pi_i : R^{\oplus n} \rightarrow R; \quad \sum_{i=1}^n r_i e_i \mapsto r_i,$$

where the elements of $R^{\oplus n}$ are written $\sum_{i=1}^n r_i e_i$ for $e_i : [n] \rightarrow R$, $e_i(k) = \begin{cases} 1, & \text{if } i = k \\ 0, & \text{else} \end{cases}$. And indeed, they span the homspace: take any $\varphi \in \text{Hom}_R(R^{\oplus n}, R)$. The identity

$$\varphi\left(\sum_{i=1}^n r_i e_i\right) = \sum_{i=1}^n r_i \varphi(e_i),$$

following from the R -linearity of φ , shows

$$\varphi = \sum_{i=1}^n \varphi(e_i) \pi_i.$$

Using this observation, one can easily verify that the map

$$\iota : [n] \rightarrow \text{Hom}_R(R^{\oplus n}, R); \quad i \mapsto \pi_i$$

satisfies the universal property for the free group on $[n]$. The conclusion is that $\text{Hom}_R(R^{\oplus n}, R) \cong R^{\oplus n}$ as R -modules. Always think about algebraic intuition before thinking about categories!!!

7.

Although A of course need not be a finite index set, for the sake of thinking of good categorical definitions of $\prod_{\alpha \in A} M_\alpha$ and $\bigoplus_{\alpha \in A} M_\alpha$, it helps to draw the respective commuting diagrams as if they were direct products/coproducts of finitely many modules. Unfortunately, I’m unsure how to typeset such a diagram concisely. Anyhow,

$$\prod_{\alpha \in A} M_\alpha := \{f : A \rightarrow \bigcup_{\alpha \in A} M_\alpha \mid \forall \alpha \in A, f(\alpha) \in M_\alpha\}$$

$$\bigoplus_{\alpha \in A} M_\alpha := \{f : A \rightarrow \bigcup_{\alpha \in A} M_\alpha \mid \forall \alpha \in A, f(\alpha) \in M_\alpha, \text{ and } f(\alpha) \neq 0 \text{ for finitely many } \alpha \in A\}$$

along with the standard projections for the product and inclusions for the direct sum satisfy the categorical universal property for product and coproduct, respectively.

Given this definition, it is clear $\mathbb{Z}^{\oplus \mathbb{Z}_+} \not\cong \mathbb{Z}^{\mathbb{Z}_+}$. There is clearly the inclusion homomorphism $\iota : \mathbb{Z}^{\oplus \mathbb{Z}_+} \hookrightarrow \mathbb{Z}^{\mathbb{Z}_+}$, but it is also clear that this map isn’t surjective.

9.

If F is the free module on A , its elements can be written as $\sum_{a \in A} r_a e_a$, where $e_a : A \rightarrow R$ is defined $e_a(a') = \begin{cases} 1, & \text{if } a' = a \\ 0, & \text{else} \end{cases}$. In other words, $\{e_a \mid a \in A\}$ form a basis for the R -module F , and it follows from the universal property of free modules that any R -module homomorphism $F \rightarrow L$ is determined entirely by how the basis elements e_a are mapped.

First, assume $\varphi : M \rightarrow N$ is surjective and take any $\alpha : F \rightarrow N$. For each $a \in A$, denote $n_a := \alpha(e_a)$. Because φ is surjective, there is $m_a \in M$ such that $\varphi(m_a) = n_a$ for each $a \in A$. Letting $\beta : F \rightarrow M$ be the map uniquely determined by the mappings $e_a \mapsto m_a$, one sees $\varphi\beta = \alpha$.

Suppose that the converse holds. Take any $n \in N$. Let $\alpha : F \rightarrow N$ be an R -module homomorphism such that $e_a \mapsto n$ for some $a \in A$ (the existence of such a homomorphism is guaranteed by the universal property of free modules). Since $\exists \beta : F \rightarrow M$ such that $\alpha = \varphi\beta$, $\varphi(\beta(e_a)) = n$. This holds for any $n \in N$, so φ is surjective.

13.

Let $A \subset M$ generate M : there is a surjective map $\iota : R^{\oplus A} \twoheadrightarrow M$. Take any R -module homomorphism $\varphi : M \rightarrow N$. Intuitively, $\varphi(A)$ generates $\text{im } \varphi$. Indeed, there are surjective mappings

$$R^{\oplus A} \xrightarrow{\iota} M \xrightarrow{\varphi} \text{im } \varphi.$$

Given any $\alpha = \sum_{a \in A} r_a a \in R^{\oplus A}$, the fact that ι and φ are R -module homomorphisms means

$$\varphi\iota(\alpha) = \sum_{a \in A} r_a \varphi(a) = \sum_{\varphi(a) \in \varphi(A)} r_{\varphi(a)} \varphi(a).$$

The last equality makes sense even when $\varphi\iota$ is not an injection on $A \subset M$; $r_{\varphi(a)}$ can be identified with the sum of all $r_{a'}$ for all $a' \in A$ such that $\varphi(a') = \varphi(a)$. The equality above shows that $\varphi\iota(R^{\oplus A})$ along with the natural injection

$$j : \varphi(A) \hookrightarrow \varphi\iota(R^{\oplus A}); \quad \varphi(a) \mapsto \varphi(a)$$

satisfies the universal property for the free module on $\varphi(A)$. Namely, the map $J : \varphi\iota(R^{\oplus A}) \rightarrow \text{im } \varphi$ induced by the inclusion map $\varphi(A) \hookrightarrow \text{im } \varphi$ is just

$$\sum_{\varphi(a) \in \varphi(A)} r_{\varphi(a)} \varphi(a) \mapsto \sum_{\varphi(a) \in \varphi(A)} r_{\varphi(a)} \varphi(a),$$

and this map must be surjective because $\varphi\iota : R^{\oplus A} \twoheadrightarrow \text{im } \varphi$ is a surjection.

14.

Denote $R = \mathbb{Z}[x_1, x_2, \dots]$ and $I := \langle x_1, x_2, \dots \rangle \triangleleft R$. Suppose I is finitely generated by $f_1, \dots, f_n \in I$. Even though $f_1, \dots, f_n$ are polynomials in infinitely many indeterminants, by definition of a polynomial, they are expressed in terms of finitely many indeterminants, say from $x_1, \dots, x_N$. Of course, by definition of I , none of $f_1, \dots, f_n$ have a constant term. $f_1, \dots, f_n$ generate I , so there are $\alpha_1, \dots, \alpha_n \in R$ such that

$$x_{N+1} = \alpha_1 f_1 + \dots + \alpha_n f_n. \tag{3}$$

However, consider the substitution map $\Phi : R \mapsto \mathbb{Z}[x_{N+1}, x_{N+2}, \dots]$ that sends $x_1, \dots, x_N$ to 0 and is constant on $\mathbb{Z}$ and all indeterminates $x_{N+1}, x_{N+2}, \dots$. Φ sends the left side of (3) to x_{N+1} but sends the right side of (3) to 0, which is a contradiction.

18.

I use bar notation for elements in the quotient M/N . Let $M/N = \langle \overline{m}_1, \dots, \overline{m}_k \rangle$ and $N = \langle n_1, \dots, n_t \rangle$. The following shows that $M = \langle m_1, \dots, m_k, n_1, \dots, n_t \rangle$. Take any $m \in M$. Letting $\pi : M \rightarrow M/N$ be the canonical projection,

$$\pi(m) = m + N = (r_1 m_1 + \dots + r_k m_k) + N$$

for some $r_1, \dots, r_k \in R$. Hence $m - (r_1 m_1 + \dots + r_k m_k) \in N$, which means

$$m = r_1 m_1 + \dots + r_k m_k + s_1 n_1 + \dots + s_t n_t$$

for some $s_1, \dots, s_t \in R$. This shows that any $m \in M$ is in the R -span of $\langle m_1, \dots, m_k, n_1, \dots, n_t \rangle$.